

Hints & Solutions :

2. eddy currents are induced hence temp. is increased hence it melt.
3. its motion is damped because magnetic flux associated with the plate keeps on changing as plate moves in & out of region between magnetic poles, flux change induces eddy currents in plates, charging of eddy current is opposite hence opposes motion of plate.
4. A force acts on it in opposite direction of motion

$$F = B \left(\frac{Bv\ell}{R} \right) \ell$$

$$M \left(\frac{dv}{dt} \right) = \frac{B^2 \ell^2 v}{R}$$

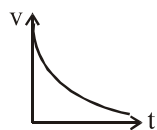
$$\int_{v_0}^v \frac{dv}{v} = \frac{B^2 \ell^2}{mR_0} \int_0^t dt$$

$$\ell n \left(\frac{v}{v_0} \right) = \frac{B^2 \ell^2}{mR} t$$

$$\frac{v}{v_0} = e^{-t/T}$$

$$v = v_0 e^{-t/T}$$

$$\text{where } T = \frac{mR}{B^2 \ell^2}$$

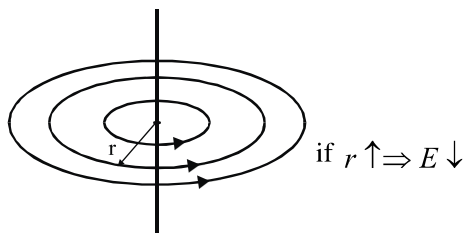


5. $W = L \int_0^I I(dI)$

$$W = \int_0^I LI^2$$

6. $\frac{B.A}{R} = \frac{Bv\ell}{v\ell R} A = \frac{IA}{v\ell} = \text{Charge}$

7. Induced electric field would be produce around circular ring as shown



9. Initially for half revolution when magnet moves towards the loop, flux increases $\Rightarrow$ clockwise current. When magnet moves away from the loop, flux decreases $\Rightarrow$ anticlockwise current
10. In a rheostat, first the insulated thin wire is doubled and then wound around the insulated cylinder. Thus, due to doubly wound wire it has a non-inductive coil with $L=0$ & effective resistance only.